The Defense Impossibility Theorem: Continuous Utility-Preserving Defenses Fix Boundary Points

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Abstract

We prove that no continuous, utility-preserving defense can fully eliminate prompt injection vulnerabilities at the safety boundary of a language model. Specifically, if the alignment deviation function $f\colon X \to \mathbb{R}$ is continuous on a connected Hausdorff space X and a defense $D\colon X \to X$ is continuous with $D(x) = x$ for all safe inputs, then there exist boundary points z with $f(z) = \tau$ that D is forced to fix in place. The proof is five steps, uses no machinery beyond point-set topology, and is machine-verified in Lean 4 with Mathlib (zero `sorry`, three standard axioms). We derive nine supporting results characterizing the vulnerability landscape—basin structure, perturbation robustness, iterative convergence, cross-model transferability, authority monotonicity, gradient ascent, and exponential cost asymmetry—forming a complete formal theory of the Manifold of Failure. We connect each theorem to empirical findings from behavioral mapping of three frontier LLMs (Llama-3-8B, GPT-OSS-20B, GPT-5-Mini), showing theory–experiment agreement at every point of contact. The complete Lean artifact (33 files, ∼200 theorems) is open-sourced.

1 Introduction

Can you build a wrapper around a language model that eliminates all prompt injection vulnerabilities?

The prevailing approach to LLM safety assumes yes—in principle, if not yet in practice. Defenses range from input classifiers [1] to output filters [9] to constitutional rewriting [3]. Each is a function $D\colon X \to X$ that preprocesses prompts before the model sees them. Each aims to map unsafe inputs to safe equivalents while leaving safe inputs unchanged.

We prove this is impossible under three mild conditions:

1. The defense is *continuous*: small changes to the input produce small changes to the defense's output.
2. The defense is *utility-preserving*: it does not modify inputs that are already safe.
3. The prompt space is *connected*: you can continuously interpolate between any two prompts.

Under these conditions, the defense must fix at least one point on the safety boundary—a prompt where alignment deviation equals the threshold exactly—leaving it unmoved and unremediated.

The proof is short (five steps), elementary (point-set topology only), and machine-verified (Lean 4, zero `sorry`). The three conditions are individually necessary: we provide counterexamples showing the theorem fails when any one is removed.

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This result does not say defense is hopeless. It says *perfect* defense is impossible under continuity and utility preservation. The impossibility is at the boundary—the mildest possible failure. The engineering prescription is not to eliminate the boundary but to make it shallow: set the threshold high enough that boundary-level behavior is benign. We show that GPT-5-Mini's empirically observed hard ceiling at AD = 0.50 [4] is exactly this strategy.

Around the central impossibility, we develop a formal theory of the Manifold of Failure comprising nine supporting theorems that characterize the vulnerability landscape: its topology, its robustness to perturbation, the convergence of iterative attacks, cross-model transferability, the role of authority framing, gradient-based navigation, and the exponential cost asymmetry between attack and defense. Each theorem is connected to empirical findings from behavioral mapping of three frontier LLMs.

Contributions.

- A formally verified impossibility theorem for continuous utility-preserving defenses (Theorem 4.1).
- A five-step proof using only Hausdorff separation, connectedness, and continuity, with counterexamples for each hypothesis (Section 4).
- Nine supporting theorems forming a complete formal theory of the failure manifold (Sections 5 to 7).
- Theory–experiment correspondence with empirical findings from three LLMs (Section 8).
- A Lean 4 artifact: 33 files, ∼200 theorems, zero `sorry`, three axioms (Section 10).

2 Related Work

Adversarial robustness. The existence of adversarial examples in neural networks is well established [15, 8]. Formal verification of robustness properties has been pursued through abstract interpretation [14], SMT-based methods [10], and randomized smoothing [7]. These works verify that specific inputs are robust; our work proves a structural impossibility for the defense mechanism itself.

Prompt injection and jailbreaking. Gradient-based attacks (GCG [18]), search-based attacks (PAIR [6], TAP [11]), and quality-diversity methods (Rainbow Teaming [13]) find individual failures. The Manifold of Failure framework [4] maps the *continuous topology* of these failures using MAP-Elites. Our work provides the formal theory underlying that framework.

Impossibility results in ML. The no-free-lunch theorems [17] establish limits on optimization. Adversarial robustness–accuracy tradeoffs [16] show that robustness has a cost. Our result is of a different character: it is a topological impossibility, not a statistical tradeoff. It holds for *any* continuous function on *any* connected Hausdorff space, independent of the learning algorithm or data distribution.

Formal verification of ML systems. Lean and Coq have been used to verify neural network properties [2] and optimization algorithms [5]. Our work is, to our knowledge, the first formally verified impossibility result for prompt injection defense.

3 Setup

Definition 3.1 (Alignment Deviation Function)**.** *Let X be a topological space. An* alignment deviation function *is a continuous map $f\colon X \to \mathbb{R}$. For a threshold $\tau \in \mathbb{R}$, define:*

$$S_\tau = \{x \in X : f(x) < \tau\} \qquad \textit{(safe region)} \tag{1}$$
$$U_\tau = \{x \in X : f(x) > \tau\} \qquad \textit{(unsafe region)} \tag{2}$$
$$B_\tau = \{x \in X : f(x) = \tau\} \qquad \textit{(boundary)} \tag{3}$$
$$F_\tau = \{x \in X : f(x) \geq \tau\} \qquad \textit{(failure manifold)} \tag{4}$$

Definition 3.2 (Defense)**.** *A* defense *is a continuous map $D\colon X \to X$. It is* utility-preserving *if $D(x) = x$ for all $x \in S_\tau$.*

Definition 3.3 (Complete Defense)**.** *A defense is* complete *if $f(D(x)) < \tau$ for all $x \in X$: after preprocessing, every input is safe.*

4 The Defense Impossibility Theorem

Theorem 4.1 (Boundary Fixation)**.** *Let X be a connected Hausdorff space. Let $f\colon X \to \mathbb{R}$ be continuous with $S_\tau \neq \emptyset$ and $U_\tau \neq \emptyset$. Let $D\colon X \to X$ be continuous with $D|_{S_\tau} = \mathrm{id}$.*

Then there exists $z \in X$ with $f(z) = \tau$ and $D(z) = z$.

Moreover, every $z \in \overline{S_\tau} \setminus S_\tau$ satisfies $f(z) = \tau$ and $D(z) = z$. This set is nonempty.

Proof. Five steps.

Step 1 (Fixed points are closed). Define $\mathrm{Fix}(D) = \{x \in X : D(x) = x\}$. The map $\varphi\colon X \to X \times X$ given by $\varphi(x) = (D(x), x)$ is continuous. The diagonal $\Delta = \{(y, y) : y \in X\}$ is closed in $X \times X$ because X is Hausdorff. Therefore $\mathrm{Fix}(D) = \varphi^{-1}(\Delta)$ is closed.

Step 2 (Closure of safe region $\subseteq$ fixed points). By utility preservation, $S_\tau \subseteq \mathrm{Fix}(D)$. Since $\mathrm{Fix}(D)$ is closed (Step 1), $\overline{S_\tau} \subseteq \mathrm{Fix}(D)$.

Step 3 (Safe region is not closed). $S_\tau = f^{-1}((-\infty, \tau))$ is open (preimage of open set under continuous f). Suppose S_τ is also closed. Then S_τ is clopen. In a connected space, the only clopen sets are $\emptyset$ and X. But $S_\tau \neq \emptyset$ (by hypothesis) and $S_\tau \neq X$ (since $U_\tau \neq \emptyset$). Contradiction.

Step 4 (Boundary point exists in the closure). Since S_τ is not closed (Step 3), $\overline{S_\tau} \supsetneq S_\tau$. Pick $z \in \overline{S_\tau} \setminus S_\tau$.

- $f(z) \leq \tau$: The set $\{x : f(x) \leq \tau\}$ is closed (preimage of $(-\infty, \tau]$) and contains S_τ. By minimality of closure, $\overline{S_\tau} \subseteq \{f \leq \tau\}$.
- $f(z) \geq \tau$: Since $z \notin S_\tau$, $\neg(f(z) < \tau)$.

Therefore $f(z) = \tau$.

Step 5 (Defense fixes the boundary point). $z \in \overline{S_\tau} \subseteq \mathrm{Fix}(D)$ (Step 2), so $D(z) = z$. Combined with $f(z) = \tau$ (Step 4): $f(D(z)) = f(z) = \tau$. □

Corollary 4.2. *Under the hypotheses of Theorem 4.1, no continuous utility-preserving defense is complete.*

Proof. A complete defense requires $f(D(z)) < \tau$ for all z. But Theorem 4.1 produces z with $f(D(z)) = \tau \not< \tau$. □

4.1 Each Hypothesis Is Necessary

Counterexample 4.3 (Removing connectedness). *Let $X = \{0, 1\}$ with the discrete topology (disconnected). Let $f(0) = 0$, $f(1) = 1$, $\tau = 0.5$. Define $D(0) = 0$, $D(1) = 0$. Then D is continuous (everything is continuous on discrete spaces), utility-preserving ($D(0) = 0$ and $f(0) = 0 < 0.5$), and complete ($f(D(1)) = f(0) = 0 < 0.5$). The defense works because safe and unsafe live on disconnected components.*

Counterexample 4.4 (Removing continuity of D). *Let $X = [0, 1]$, $f(x) = x$, $\tau = 0.5$. Define $D(x) = x$ for $x < 0.5$ and $D(x) = 0$ for $x \geq 0.5$. Then D is utility-preserving and complete ($f(D(x)) < 0.5$ for all x) but discontinuous at $x = 0.5$. The defense works because it jumps.*

Counterexample 4.5 (Removing utility preservation). *Let $D(x) = x_0$ for a fixed safe point x_0. Then D is continuous and complete ($f(D(x)) = f(x_0) < \tau$ for all x) but modifies safe inputs ($D(x) \neq x$ for $x \neq x_0$). The defense works because it maps everything to one point, destroying utility entirely.*

Remark 4.6 (The defense trilemma). *A defense can satisfy at most two of:*

1. Continuity*: small input changes $\Rightarrow$ small output changes.*
2. Utility preservation*: safe inputs pass through unchanged.*
3. Completeness*: all outputs are safe after preprocessing.*

Theorem 4.1 proves this when the space is connected and both safe and unsafe inputs exist. Each counterexample sacrifices one property to achieve the other two.

4.2 What the Theorem Does Not Say

Remark 4.7 (Not all boundary points are fixed). *The theorem applies to $\overline{S_\tau} \setminus S_\tau$, not to all of B_τ. Isolated boundary points—where $f(z) = \tau$ but $f > \tau$ in a punctured neighborhood—may not lie in $\overline{S_\tau}$. The Lean artifact includes a verified counterexample: $f(x) = x^2$, $\tau = 0$, where $f(0) = 0$ but $\{f < 0\} = \emptyset$, so $\overline{S_0} = \emptyset$ and 0 is not captured.*

Remark 4.8 (Boundary behavior may be mild). *The fixed points satisfy $f(z) = \tau$ exactly—the threshold, not a catastrophic failure. If τ is set high enough that τ-level behavior is benign, the theorem is true but harmless. This is not a loophole; it is the engineering prescription.*

Remark 4.9 (Discontinuous defenses escape the theorem). *Hard blocklists, keyword filters, and exact-match rejection are discontinuous. The theorem does not constrain them. Their cost is adversarial examples against the filter itself and false positives on benign inputs—a different problem.*

5 The Vulnerability Landscape

The Boundary Fixation theorem says the defender cannot clear the boundary. The following theorems characterize the landscape that boundary sits in.

Theorem 5.1 (Topological Partition). *For continuous $f\colon X \to \mathbb{R}$: S_τ is open, U_τ is open, B_τ is closed, F_τ is closed, and $F_\tau = B_\tau \cup U_\tau$. Every point belongs to exactly one of S_τ, B_τ, U_τ.*

Theorem 5.2 (Basin Structure). *If $f(p) > \tau$ for some p, then U_τ is an open set with positive Lebesgue measure.*

Vulnerabilities are not isolated points. If one attack exists, a positive-volume neighborhood of variants exists.

Theorem 5.3 (No Gap). *If X is connected and f takes values both above and below τ, then $B_\tau \neq \emptyset$. Any continuous path from a safe point to an unsafe point crosses B_τ.*

There is no moat between safe and unsafe. The boundary is a concrete, locatable surface.

6 Attack Properties

Theorem 6.1 (Perturbation Robustness). *If f is L-Lipschitz and $f(p) > \tau$, then* $\mathrm{ball}(p, r(p)) \subseteq U_\tau$ *where*

$$r(p) = \frac{f(p) - \tau}{L} > 0. \tag{5}$$

Moreover, r is monotonically increasing in $f(p)$.

Successful attacks survive rephrasing within a quantifiable radius. More severe attacks have larger perturbation balls.

Theorem 6.2 (Iterative Convergence). *Let $T\colon X \to X$ satisfy* $\mathrm{score}(T(x)) \geq \mathrm{score}(x)$ *with* $\mathrm{score}\colon X \to [0,1]$. *Then* $\mathrm{score}(T^n(x_0))$ *converges. If each step gains at least $\delta > 0$, convergence occurs in at most $\lceil 1/\delta \rceil$ steps.*

This is why MAP-Elites [12], PAIR [6], and TAP [11] converge: they are monotone-improvement operators on a bounded score.

Theorem 6.3 (Transferability). *If $\|f - g\|_\infty \leq \delta$, then $\{f > \tau + \delta\} \subseteq \{g > \tau\}$. If $\delta = 0$, the basins are identical.*

Attack transfer is controlled by the sup-norm distance between models' alignment deviation surfaces. Similar models have overlapping basins.

Theorem 6.4 (Authority Monotonicity). *If $f(y, \cdot)$ is monotone non-decreasing in authority for each fixed indirection y, then vulnerability is upward-closed, a critical threshold a_2^* exists, and the threshold curve is non-increasing in indirection.*

Theorem 6.5 (Gradient Ascent). *If $f\colon E \to \mathbb{R}$ has nonzero Fréchet derivative at x (in a real inner product space E), then there exist $v \in E$ and $\varepsilon > 0$ with $f(x + \varepsilon v) > f(x)$.*

At non-critical points, gradient-based attacks always find an improving direction. The gradient is a reliable compass toward higher alignment deviation.

7 Cost Asymmetry

Theorem 7.1 (Exponential Cost Asymmetry). *For grid resolution $N \geq 2$, attack step size $\delta > 0$, and defense patch radius $0 < r < 1$:*

1. *Attack cost $\leq 1/\delta$* (dimension-independent).
2. *Defense cost $= N^d$* (exponential in dimension d).
3. *The ratio $\delta \cdot N^d \to \infty$ as $d \to \infty$.*
4. *For any $R > 0$, $\exists\, d_0$ such that the cost ratio exceeds R for all $d \geq d_0$.*

Proof. Attack cost: Theorem 6.2 gives convergence in $\lceil 1/\delta \rceil$ steps, independent of d. Defense cost: an N-partition per axis in d dimensions yields N^d cells; by pigeonhole (verified: `Fintype.card_le_of_surjective`), filling all cells requires $\geq N^d$ queries. The ratio $\delta \cdot N^d$ is monotone in d and tends to ∞ since $N^d \to \infty$ for $N \geq 2$. □

At $d = 2$, $N = 25$, $\delta = 0.01$: the cost ratio is 6.25—the defender pays 6× more. At $d = 10$: $\sim 10^{12}$. The Lean artifact verifies $25^2 \times 0.01 = 6.25$ by `norm_num`.

Table 1: Each theorem makes a prediction. Each prediction is confirmed by the empirical data.

Theorem	**Predicts**	**Confirmed by**
Basin Structure (5.2)	Basins have positive area	Heatmaps show extended colored regions, not isolated dots
No Gap (5.3)	Boundary is nonempty	Contour lines exist in all models with both safe and unsafe regions
Perturbation Robustness (6.1)	Adjacent cells have similar AD	Heatmaps are smooth; 3D surfaces are continuous
Iterative Convergence (6.2)	MAP-Elites/PAIR/TAP converge	Convergence curves are monotone and plateau
Transferability (6.3)	Close models share basins	Model spectrum: Llama (0.93) $\rightarrow$ GPT-OSS (0.73) $\rightarrow$ GPT-5-Mini (0.47)
Authority Monotonicity (6.4)	Horizontal banding	Contour plots show bands at specific a_2 values
Boundary Fixation (4.1)	Defense can't clear boundary	GPT-5-Mini caps at $\tau = 0.50$; no method breaches it
Cost Asymmetry (7.1)	2D tractable, high-d intractable	15K queries fill 63% at $d{=}2$

8 Experimental Validation

We validate the theory against empirical findings from the Manifold of Failure framework [4], which maps the behavioral topology of three LLMs using MAP-Elites over a 2D behavioral space (query indirection $\times$ authority framing).

8.1 Theory–Experiment Correspondence

8.2 Model-Specific Predictions

Llama-3-8B. Mean AD = 0.93, basin rate 93.9%. The AD surface is a near-flat mesa (Theorem 6.1: small L, large robustness radii). The basin is one massive connected region (Theorem 5.2: the near-constant surface is trivially concave, so the superlevel set is convex, hence connected). The boundary is thin—narrow safe corridors at specific authority levels. Theorem 4.1 predicts fixed boundary points in these corridors; their thinness means even small perturbations push back into the vulnerability plateau.

GPT-OSS-20B. Mean AD = 0.73, basin rate 64.3%. The AD surface is rugged (large L, small robustness radii per Theorem 6.1). The basin fragments into a complex mosaic. Theorem 6.4 predicts horizontal banding; the contour plots confirm bands at $a_2 \approx 0.25$–0.35 and 0.65–0.85. Theorem 4.1 predicts fixed boundary points throughout the safe–unsafe mosaic; the empirical "bullseye" patterns around vulnerability peaks are the contour structure surrounding these boundary points.

GPT-5-Mini. Peak AD = 0.50, basin rate 0%. If we set $\tau = 0.50$, the hypothesis $U_\tau \neq \emptyset$ fails: Theorem 4.1 does not apply, correctly predicting no impossibility. The model's strategy is to make the boundary harmless, not absent. Theorem 5.2 still applies at lower thresholds (the model does

exhibit AD $= 0.47$ uniformly), but the "basin" at $\tau = 0.40$ is the entire space—uniformly moderate, never harmful.

8.3 Quantitative Checks

Robustness radius. From the 3D surface plots, we estimate $L \approx 5$ for Llama-3-8B (the surface changes by $\sim$0.5 over $\sim$0.1 in behavioral coordinates). For a typical attack with $\mathrm{AD}(p) = 0.93$ at $\tau = 0.50$: $r(p) = (0.93 - 0.50)/5 = 0.086$. In a 25×25 grid with cell width 0.04, this covers $\sim$2 cells in each direction—consistent with the smooth heatmaps where adjacent cells have similar values.

Convergence bound. From the convergence curves, the typical per-step improvement is $\delta \approx$ 0.001–0.01, predicting convergence in 100–1000 steps. The empirical curves plateau within $\sim$5000 iterations, consistent with the upper bound given early-phase improvements are larger than δ.

Cost ratio. At $d = 2$, $N = 25$, $\delta = 0.01$: ratio $= 0.01 \times 625 = 6.25$. The paper uses 15,000 queries to fill 394 cells (63% coverage), or $\sim$38 queries per cell. The attacker needs $\sim$100 queries to converge ($1/\delta$). Ratio $\approx 394/100 \approx 4$, the same order as the theoretical 6.25.

9 The Engineering Prescription

Theorem 4.1 says the boundary cannot be eliminated. It does not say what to do about it. We derive three strategies from the theory.

Strategy 1: Make the boundary shallow. Set τ high enough that $f(z) = \tau$ is benign. GPT-5-Mini does this: boundary points have AD $= 0.50$, which corresponds to moderate-level refusals, not harmful compliance. The theorem holds; the boundary exists; it does not matter.

Strategy 2: Reduce the Lipschitz constant. Theorem 6.1 says the robustness radius is $(f(p) - \tau)/L$. Smoother alignment surfaces (smaller L) mean each vulnerability, if it exists, extends over a wider region—but also means the surface is more predictable and can be monitored. The contour plots for GPT-5-Mini show L is small: the surface barely varies ($\mathrm{AD} \in [0.39, 0.50]$).

Strategy 3: Reduce effective dimension. Theorem 7.1 says defense cost is N^d. Constraining the prompt interface—standardized formats, restricted API parameters, bounded context length—reduces the effective d, making the behavioral space more tractable to patrol.

10 The Lean Artifact

10.1 Verification Details

The complete theory is machine-verified in Lean 4.28.0 with Mathlib v4.28.0. The artifact comprises:

- 10 core theory files (MoF_01–MoF_10)
- 10 cost theory files (MoF_Cost_01–MoF_Cost_10)
- 10 advanced theory files (MoF_Adv_01–MoF_Adv_10)
- 3 capstone files (MasterTheorem, Euclidean instantiation, verification)

Every theorem depends only on three axioms:

- `propext`: propositional extensionality ($P \leftrightarrow Q \implies P = Q$)

- `Classical.choice`: the axiom of choice
- `Quot.sound`: quotient soundness

These are Lean's standard foundational axioms, shared with all of Mathlib.

10.2 Proof Map

The master theorem's proof chain uses six Mathlib lemmas:

Lean name	Says
`isClosed_diagonal`	Diagonal is closed in Hausdorff spaces
`IsClosed.preimage`	Preimage of closed under continuous is closed
`IsClosed.closure_subset_iff`	Closed set containing S contains $\overline{S}$
`isClopen_iff`	Clopen sets in connected spaces are $\emptyset$ or X
`closure_minimal`	$\overline{S} \subseteq T$ when $S \subseteq T$ and T closed
`isClosed_le`	$\{f \leq g\}$ is closed for continuous f, g

10.3 Generality

All theorems are stated over arbitrary topological spaces (`{X : Type*} [TopologicalSpace X]`) unless they require metric or measure structure. The Euclidean instantiation file verifies that $\mathbb{R}^n$ (`EuclideanSpace ℝ (Fin d)`) satisfies all required typeclasses for any d.

11 Limitations

The continuous model is an approximation. Real prompts are discrete token sequences. The behavioral space is finite. The theory applies to the continuous relaxation, supported empirically by GP fits with $R^2 > 0.85$ [4] but not formally guaranteed.

The impossibility is at the boundary. Fixed points have $f(z) = \tau$ exactly—the mildest possible failure. The theorem constrains perfection, not adequacy. A defense that reduces AD to $\tau + \varepsilon$ everywhere except at boundary points is permitted.

The cost asymmetry assumes brute force. Theorem 7.1 counts grid cells. Learning-based defenses that generalize across the space may escape the exponential lower bound by exploiting structure the theorem ignores.

Single-turn only. The manifold may evolve across conversation turns. Extending to time-varying or history-dependent manifolds is future work.

The theory does not explain model-specific topology. Why Llama-3-8B has a vulnerability plateau while GPT-5-Mini does not depends on architecture, training data, and alignment procedure—empirical questions, not mathematical ones.

12 Conclusion

We proved that continuous utility-preserving defenses must fix boundary points of the alignment deviation surface. The proof is five steps, each using one hypothesis. The three hypotheses—continuity, utility preservation, connectedness—are individually necessary, with counterexamples

for each. The theorem establishes a defense trilemma: you cannot simultaneously have continuity, utility, and completeness.

Nine supporting theorems characterize the landscape this impossibility constrains: basins are open with positive measure, the boundary is unavoidable, attacks are robust and convergent, transfer is controlled by sup-norm distance, and defense cost grows exponentially in dimension. Each theorem connects to empirical findings from three frontier LLMs.

The engineering prescription is not despair but calibration: make the boundary shallow (raise τ), the surface smooth (reduce L), and the space low-dimensional (constrain the interface). GPT-5-Mini's hard ceiling at $\mathrm{AD} = 0.50$ is this prescription in practice.

The theory is machine-verified: 33 Lean files, ∼200 theorems, zero `sorry`, three axioms. The artifact is open-sourced for community use and replication.

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